WIMP51 - 8 bit architecture

PC - program Counter
IR - instruction registe

R0-R7 general registers

AUX - auxiliary register ⇒ intermediate/
temporary data

ACC - accumulator - general register
for data & storing results

PC_ALU ⇒ used to update PC

ALU ⇒ arithmetic logic unit
C, Z carry, zero

CU - control Unit ⇒ quarterback
provides signals to coordinate
program instruction execution

we ⇒ write enables for each register

$\overline{PSEN}$ - program store enable
⇒ connected to OE (output
Enable)

Draw the timing diagram for the following control signals on the diagram below when the WIMP51 is executing the instruction MOV A, 12h Label the fetch, decode and execute parts of the instruction cycle on the clock portion of the signal below.

0111 0100 0001 0010 Code memory

The dotted lines have been drawn as a graphical reference only and do not reflect the value of the corresponding signal. If you think a particular control signal is of no significance during a particular time period, indicate this opinion by using a **dashed** line to denote the signal as a "don't care" for that time period. Give "command" values for pcalu_op and alu_op; for example, indicate the value of "pcalu_op" as "PC_INC" or "PC_ADD" rather than representing this command by a numeric code. If you are not sure about a particular control signal, make an educated guess, indicate that the value you have marked is a guess, and explain why you made this particular guess. Remember, PSEN/ pulses low whenever a byte is being fetched from code memory.

The WIMP51 architecture schematic is available in the notes provided to you for this exam.

